

One number tells
you how much
energy a data centre
wastes.



Power Usage Effectiveness

Power Usage *Effectiveness.*

It compares the energy a whole data centre uses with the energy that actually reaches the computers.

PUE =

Total facility energy

computers + cooling + lighting + power losses

IT equipment energy

servers, storage and network only

A perfect score is **1.0**: every unit of power goes to computing.

Lower is better.
Much better.

2.0

Computing

Overhead

1.5

Computing

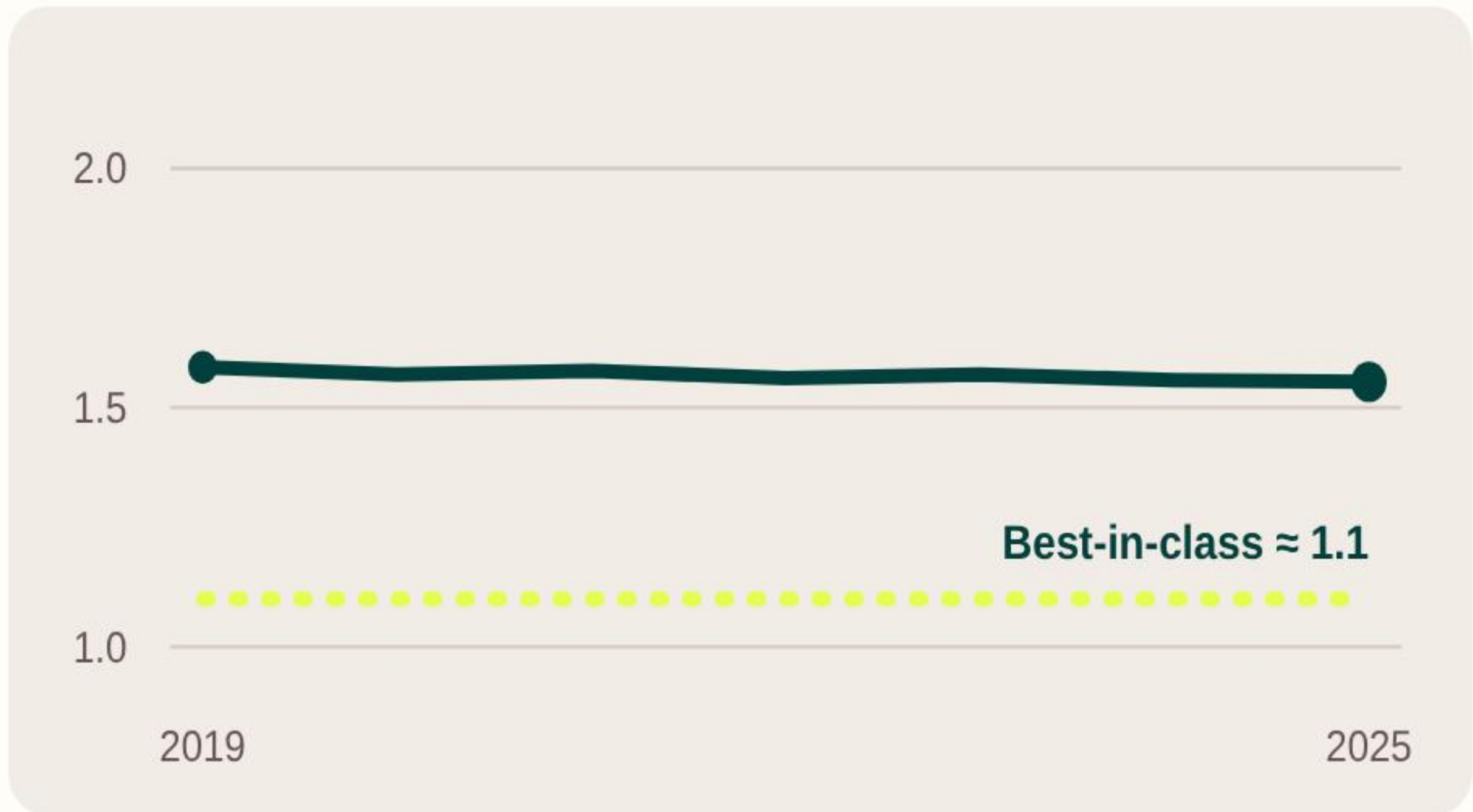
Overhead

1.1

Computing

At **2.0**, every unit of power for computing needs **one more** for cooling and losses. At **1.1**, it needs just **a tenth**.

The world average is *stuck above 1.5.*



Average PUE has barely moved in **six years**, held back by older buildings and hot climates. The most efficient large facilities report figures close to 1.1.

Source: Uptime Institute, Global Data Center Survey 2025 (weighted average PUE above 1.5). Chart is illustrative of the flat trend.

Heat and humidity make it *harder here.*



Hot summers

Cooling systems work harder for more of the year



Humid coasts

Mumbai and Chennai, India's biggest hubs, are coastal and humid



So design matters more

Smart engineering can close the gap with cooler countries

Four proven ways to *bring PUE down.*



Contain the airflow

Separate hot and cold aisles so cool air isn't wasted



Run a little warmer

Raise temperature setpoints within safe equipment limits



Cool with liquid

Direct liquid cooling handles dense AI racks efficiently



Cut power losses

Efficient UPS and power systems waste less energy

— THE HONEST CAVEAT

PUE isn't the *whole story.*

A data centre can be very efficient and still run on coal power, or use a lot of water. Look at three numbers together.

PUE

Energy efficiency

How much power is overhead

WUE

Water efficiency

Litres of water per unit of IT energy

RE%

Renewable share

How much of the electricity is clean

Put these numbers *in every contract.*

✓ Annual PUE, measured, not designed

✓ Water use and water source

✓ Share of renewable electricity

✓ A plan to improve all three

— THE TAKEAWAY

What gets measured
gets improved.
Start with PUE.
Don't stop there.

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